

PRACTICE SHEET-1

1. A train started from a station with a certain number of passengers. At the first halt, $\frac{1}{3}$ rd of its passengers got down and 120 passengers got in. At the second halt, half of the passengers got down and 100 persons got in. Then, the train left for its destination with 240 passengers. How many passengers were there in the train when it started?
(a) 540 (b) 480
(c) 360 (d) 240
2. A person bought a certain number of books for Rs 80. If he had bought 4 more books for the same sum, each book would have cost Rs 1 less. What is the price of each book?
(a) Rs 10 (b) Rs 8
(c) Rs 5 (d) Rs 4
3. A person bought 5 tickets from a station P to a station Q and 10 tickets from the station P to a station R . He paid Rs 350. If the sum of a ticket from P to Q and a ticket from P to R is Rs 42, then what is the fare from P to Q ?
(a) Rs 12 (b) Rs 14
(c) Rs 16 (d) Rs 18
4. Pooja started her job with certain monthly salary and gets a fixed increment every year. If her salary was Rs 4200 after 3 years and Rs 6800 after 8 years of service, then what are her initial salary and the annual increment, respectively?
(a) Rs 2640, Rs 320 (b) Rs 2460, Rs 320
(c) Rs 2460, Rs 520 (d) Rs 2640, Rs 520
5. A number consists of two digits, whose sum is 10. If 18 is subtracted from the number, digits of the number are reversed. What is the product?
(a) 15 (b) 18
(c) 24 (d) 32
6. A railway ticket for a child costs half the full fare but the reservation charge is the same on half tickets as much as on full ticket. One reserved first class ticket for a journey between two stations is Rs 362, one full and one half reserved first class tickets cost Rs 554. What is the reservation charge?
(a) Rs 18 (b) Rs 22
(c) Rs 38 (d) Rs 46
7. Let there be three simultaneous linear equations in two unknowns, which are non-parallel and non-collinear them. What can be the number of solutions (if they do exist)?
(a) One or infinite (b) Only one
(c) Exactly two (d) exactly three
8. If 1 is added to the denominator of a fraction, it becomes $\frac{1}{2}$ and if 1 is added to the numerator, the fraction becomes 1. What is the fraction?
(a) $\frac{5}{9}$ (b) $\frac{2}{3}$
(c) $\frac{4}{7}$ (d) $\frac{10}{11}$
9. If $\frac{2}{x} + \frac{3}{y} = \frac{9}{xy}$ and $\frac{4}{x} + \frac{9}{y} = \frac{21}{xy}$, where $x \neq 0$ and $y \neq 0$, then what is the value of $x + y$?
(a) 2 (b) 3
(c) 4 (d) 8
10. The sum of two numbers is 80. If the larger number exceeds four times the smaller by 5, what is the smaller number?
(a) 5 (b) 15
(c) 20 (d) 25
11. If $2^a + 3^b = 17$ and $2^{a+2} - 3^{b+1} = 5$, then,
(a) $a = 2, b = 3$ (b) $a = -2, b = 3$
(c) $a = 2, b = -3$ (d) $a = 3, b = 2$
12. If $\frac{2x-3y+1}{2} = \frac{x+4y+8}{3} = \frac{4x-7y+2}{5}$ then what is $(x + y)$ equal to?
(a) 3 (b) 2
(c) 0 (d) -2
13. Sum of two numbers is 21 and their difference is 11, then the greatest number is
(a) 5 (b) 16
(c) 9 (d) 10
14. The area of a rectangle remains the same if the length is increased by 7 m and the breadth is decreased by 3 m. The area remains unaffected if the length is decreased by 7 m and the breadth is increased by 5 m, then area of rectangle is
(a) 280 m^2 (b) 320 m^2
(c) 420 m^2 (d) 400 m^2

- 15.** A and B each have a certain number of mangoes. A says to B: "If you give 30 of your mangoes, I will have twice as many as left with you" B replies "If you give me 10, I will have thrice as many as left with you". How many mangoes did A has?
(a) 41 (b) 62
(c) 34 (d) 32
- 16.** There are two examination rooms A and B. If 10 candidates are sent from room A to room B the number of students in each room is the same. If 20 candidates are sent from B to A, the number of students in A is double the number in of students in B. Then, number of students room B is
(a) 40 (b) 100
(c) 80 (d) 60
- 17.** The numerator of a fraction is 2 less than the denominator. If one is added to its denominator, it becomes $\frac{1}{2}$, the fraction is
(a) $\frac{2}{3}$ (b) $\frac{3}{5}$
(c) $\frac{3}{4}$ (d) $\frac{5}{7}$
- 18.** The sum of digits of a two-digit number is 8 and the difference between the number and that formed by reversing the digits is 18. What is three difference between the digits of the number?
(a) 1 (b) 2
(c) 3 (d) 4
- 19.** A length of rectangle is 5 cm less than twice its width. if the length is decreased by 5 cm, and width is decreased by 2 cm. the perimeter of resulting rectangle will be 18 cm, Find the length of original rectangle ?
(a) 9 (b) 7
(c) 6 (d) 5
- 20.** The present of Sita's father is three times the present age of Sita. After six years sum of their ages will be 72 years. Find the present age of Sita ?
(a) 12 yrs (b) 15 yrs
(c) 16 yrs (d) 10 yrs
- 21.** The Community Relief fund receives a large donation of \$ 2800. The foundation agrees to spend the money on \$ 20 school bags, \$ 25 sweaters, \$ 5 b0oks. They want to buy 200 items and send them to schools in earthquake-hit areas. They must order as many books as school bages and sweaters combined. How many of each item should they order?
(a) (40, 60, 100) (b) (20, 30, 80)
(c) (50, 100, 60) (d) (40, 80, 25)
- 22.** A student was asked to find $\frac{5}{16}$ of a number. By mistake he found $\frac{5}{6}$ of that number. His answer was 250 more than the correct answer. Find the given number.
(1) 300 (2) 480
(3) 450 (4) 500
- 23.** A tin of oil was $\frac{4}{5}$ full. When 6 bottles of oil was taken out and 4 bottles of oil was poured into it, it was $\frac{3}{4}$ full. How many bottles of oil can the tin contain?
(1) 10 (2) 20
(3) 30 (4) 40
- 24.** A candidate in an examination was asked to find $\frac{5}{14}$ of a certain number. By mistake he found $\frac{5}{4}$ of it. Thus, his answer was 25 more than the correct answer. The number was:
(1) 28 (2) 56
(3) 84 (4) 140
- 25.** A man read $\frac{2}{5}$ th of a book on the first day. He read $\frac{1}{3}$ rd more on second day than he read on the first day. 15 pages were left for the third day. The number of pages in the book is
(1) 100 (2) 105
(3) 225 (4) 250
- 26.** A girl was asked to multiply a number by $\frac{7}{8}$, instead she divided the number by $\frac{7}{8}$ and got the result 15 more than the correct result. The sum of the digits of the number was:
(1) 4 (2) 8
(3) 6 (4) 11

- 27.** A, B, C and D purchase a gift worth Rs 60. A pays $\frac{1}{2}$ of what others are paying, B pays $\frac{1}{3}$ of what others are paying and C pays $\frac{1}{4}$ of what others are paying. What is the amount paid by D?
 (1) 16 (2) 13
 (3) 14 (4) 15
- 28.** A number of boys raised 400 for a famine relief fund, each boy giving as many 25 paise coins as there were boys. The number of boys was:
 (1) 40 (2) 16
 (3) 20 (4) 100
- 29.** Thrice the square of a natural number decreased by four times the number is equal to 50 more than the number. The number is:
 (1) 4 (2) 5
 (3) 10 (4) 6
- 30.** A farmer divides his herd of n cows among his four sons so that the first son gets one – half the herd, the second son gets one – fourth, the third son gets one – fifth and the fourth son gets 7 cows. The value of n is
 (1) 80 (2) 100
 (3) 140 (4) 180

ANSWER KEY

1.	(d)	2.	(c)	3.	(b)	4.	(d)	5.	(c)	6.	(b)	7.	(b)	8.	(b)	9.	(c)	10.	(b)
11.	(d)	12.	(d)	13.	(b)	14.	(c)	15.	(c)	16.	(c)	17.	(b)	18.	(b)	19.	(a)	20.	(b)
21.	(a)	22.	(b)	23.	(d)	24.	(a)	25.	(c)	26.	(d)	27.	(b)	28.	(a)	29.	(b)	30.	(c)

PRACTICE SHEET-2

1. A man has some hens and cows. If the number of heads be 48 and the number of feet equals 140, then the number of hens will be
 (1) 23 (2) 22
 (3) 24 (4) 26
2. Slope of the line $2x - y + 4 = 0$ is
 (1) 2 (2) 1
 (3) 0 (4) $1/2$
3. A number consists of two digits. If the number formed by interchanging the digits is added to the original number, the resulting number (i.e. the sum) must be divisible by
 (1) 11 (2) 9
 (3) 5 (4) 3
4. If 1 is added to the denominator of a fraction it becomes $\frac{1}{2}$. If 1 is added to the numerator it becomes 1. The product of numerator and denominator of the fraction is
 (1) 6 (2) 10
 (3) 12 (4) 14
5. A fraction becomes $\frac{1}{3}$ when 1 is subtracted from both the numerator and the denominator. The same fraction becomes $\frac{1}{2}$ when 1 is added to both the numerator and the denominator. The sum of numerator and denominator of the fraction is
 (1) 10 (2) 18
 (3) 7 (4) 16
6. If 1 is added to both the numerator and the denominator of a fraction, it becomes $\frac{1}{4}$. If 2 is added to both the numerator and the denominator of that fraction, it becomes $\frac{1}{3}$. The sum of numerator and denominator of the fraction is:
 (1) 8 (2) 13
 (3) 22 (4) 27
7. In a two digit number if it is known that its units digit exceeds its tens digit by 2 and that the product of the given number and the sum of its digits is equal to 144, then the number is
 (1) 46 (2) 42
 (3) 26 (4) 24
8. In a test, 1 mark is awarded for each correct answer and one mark is deducted for each wrong answer. If a boy answers all 20 items of the test and gets 8 marks, the number of questions answered correct by him was
 (1) 16 (2) 14
 (3) 12 (4) 8
9. The difference between two positive numbers is 3. If the sum of their squares is 369, then the sum of the numbers is:
 (1) 81 (2) 33
 (3) 27 (4) 25
10. A number consists of two digits such that the digit in the ten's place is less by 2 than the digit in the unit's place. Three times the number added to $\frac{6}{7}$ times the number obtained by reversing the digits equals 108. The sum of digits in the number is:
 (1) 8 (2) 9
 (3) 6 (4) 7
11. Of the three numbers, the second is twice the first and it is also thrice the third. If the average of three numbers is 44, the difference of the first number and the third number is:
 (1) 24 (2) 18
 (3) 12 (4) 6
12. A two digit number is five times the sum of its digits. If 9 is added to the number, the digits interchange their positions. The sum of digits of the number is:
 (1) 11 (2) 9
 (3) 7 (4) 6
13. Solution of equation $x + y = 5$ and $2x + y - 6 = 0$ is
 (1) $x = 1, y = 2$ (2) $x = 1, y = 4$
 (3) $x = 2, y = 3$ (4) $x = 4, y = 2$

- 14.** Straight line $2x - 3y - 6 = 0$ will cut the line $4x + ky = 15$ if
 (1) $k = 1$ (2) $k = 2$
 (3) $k = 3$ (4) All of above
- 15.** The product of two numbers is 24 times the difference of these two numbers. If the sum of these numbers is 14, the larger number is
 (1) 9 (2) 8
 (3) 7 (4) 10
- 16.** If a and b are negative real numbers and c is a positive real number, then which of the following is/are correct?
 1. $a - b < a - c$
 2. If $a < b$ then $\frac{a}{c} < \frac{b}{c}$
 3. $\frac{1}{b} < \frac{1}{c}$
 Select the correct answer using the code given below:
 (1) 1 (2) 2 only
 (3) 3 only (4) 2 and 3
- 17.** Equation $x + 3y - 6 = 0$ will cut the Y axis
 (1) 6 unit above the origin
 (2) 2 unit above the origin
 (3) 6 unit below the origin
 (4) 2 unit below the origin
- 18.** The graph of equation $y = 5$ is parallel to
 (1) Y axis (2) X axis
 (3) Both axis (4) None of these
- 19.** Which of the following is not a method to solving linear equation of two variables?
 (1) Elimination method
 (2) Substitution method
 (3) Cross multiplication method
 (4) Division method
- 20.** Which of the following is not a linear equation?
 (1) $ax + by + c = 0$
 (2) $ax + c = 0$
 (3) $ax + by + cz + d = 0$
 (4) $(ax + b)x + c = 0$
- 21.** In an examination, a student scores 4 marks for every correct answer and loses 1 mark for every wrong answer. A student attempted all the 200 questions and scored in all 200 marks. The number of questions, he answered correctly was
 (1) 82 (2) 80
 (3) 68 (4) 60
- 22.** What is/are the solutions of the set of homogeneous equations $(4x + 2y = 0)$ and $(6x + 3y = 0)$?
 (a) Only $x = 0, y = 0$
 (b) Only $x = 0, y = 0$ and $x = 1, y = 2$
 (c) An infinite number of solutions
 (d) No solution
- 23.** The cost of 4 books and 3 pencils is same as that of 8 books and 1 pencil. This cost will be same as that of which one of the following?
 (a) 2 books and 6 pencils
 (b) 5 books and 5 pencils
 (c) 6 books and 2 pencils
 (d) 12 books and 4 pencils
- 24.** If one-third of a two-digit number exceeds its one-fourth by 8, then what is the sum of the digits of the number?
 (a) 6 (b) 13
 (c) 15 (d) 17
- 25.** What is the sum of two numbers whose difference is 45 and the quotient of the greater number by the lesser number is 4?
 (a) 100 (b) 90
 (c) 80 (d) 75
- 26.** The solution of the equations $\frac{3x - y + 1}{3} = \frac{2x + y + 2}{5} = \frac{3x + 2y + 1}{6}$ is given by which one of the following?
 (a) $x = 2, y = 1$
 (b) $x = 1, y = 1$
 (c) $x = -1, y = -1$
 (d) $x = 1, y = 2$
- 27.** What is the solution of the equations $x - y = 0.9$ and $11(x + y)^{-1} = 2$?
 (a) $x = 3.2$ and $y = 2.3$
 (b) $x = 1$ and $y = 0.1$
 (c) $x = 2$ and $y = 1.1$
 (d) $x = 1.2$ and $y = 0.3$
- 28.** What is the value of k for which the system of equations $x + 2y - 3 = 0$ and $5x + ky + 7 = 0$ has no solution?
 (a) $-3/14$ (b) $-14/3$
 (c) $1/10$ (d) 10

29. Under what condition do the equations $kx - y = 2$ and $6x - 2y = 3$ have a unique solution?

- (a) $k = 3$ (b) $k \neq 3$
(c) $k = 0$ (d) $k \neq 0$

30. For what value of k , the following equation will be inconsistent? $2x - ky = 4$ and $3x + 2y = 6$?

- (a) $4/3$ (b) $-4/3$
(c) $2/3$ (d) $3/2$

ANSWER KEY

1.	(4)	2.	(1)	3.	(1)	4.	(1)	5.	(1)	6.	(1)	7.	(4)	8.	(2)	9.	(3)	10.	(3)
11.	(3)	12.	(2)	13.	(2)	14.	(4)	15.	(2)	16.	(4)	17.	(2)	18.	(2)	19.	(4)	20.	(4)
21.	(2)	22.	(3)	23.	(3)	24.	(3)	25.	(4)	26.	(2)	27.	(1)	28.	(4)	29.	(2)	30.	(2)

PRACTICE SHEET-3

1. Every quadratic equation $ax^2 + bx + c = 0$, where $a, b, c \in \mathbb{R}$, $a \neq 0$ has:

- (a) Exactly one real root
(b) At least one real root
(c) At least two real roots
(d) At most two real roots

2. If α, β are roots of $ax^2 + bx + c = 0$ and $\alpha + h, \beta + h$ are the roots of $px^2 + qx + r = 0$, then what is h equal to?

- (a) $\frac{1}{2} \left(\frac{b}{a} - \frac{q}{p} \right)$ (b) $\frac{1}{2} \left(-\frac{b}{a} + \frac{q}{p} \right)$
(c) $\frac{1}{2} \left(\frac{b}{p} + \frac{q}{a} \right)$ (d) $\frac{1}{2} \left(-\frac{b}{a} + \frac{q}{a} \right)$

3. If α and β are the roots the equation $ax^2 + bx + c = 0$, where $a \neq 0$, then $(a\alpha + b)(a\beta + b)$ is equal to:

- (a) ab (b) bc
(c) ca (d) abc

4. The roots of the equation $2a^2 x^2 - 2abx + b^2 = 0$, when $a < 0$ and $b > 0$ are:

- (a) Sometimes complex (b) Always irrational
(c) Always complex (d) Always real

5. The quadratic equation $x^2 + bx + 4 = 0$ will have real roots, if:

- (a) Only $b \leq -4$ (b) Only $b \geq 4$
(c) $-4 < b < 4$ (d) $b \leq -4, b \geq 4$

6. If α and β are the roots of the equation $x^2 + x + 2 = 0$, then what is $\frac{\alpha^{10} + \beta^{10}}{\alpha^{-10} + \beta^{-10}}$ equal to?

- (a) 4096 (b) 2048
(c) 1024 (d) 512

7. What is the difference in the roots of the equation $x^2 - 10x + 9 = 0$.

- (a) 2 (b) 3
(c) 5 (d) 8

8. If α and β are the roots of the equation $ax^2 + bx + c = 0$, then what is the value of $\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} + \sqrt{\frac{b}{a}} = ?$

- (a) -10 (b) 0
(c) 1 (d) 2

9. The roots of the equation $x^2 - 8x + 16 = 0$:

- (a) Are imaginary
(b) Are distinct and real
(c) Are equal and real
(d) Cannot be determined

10. If a and b are rational and b is not perfect square, then the quadratic equation with rational coefficients whose one root is $3a + \sqrt{b}$ is:

- (a) $x^2 - 6ax + 9a^2 - b = 0$
(b) $3ax^2 + x - \sqrt{b} = 0$
(c) $x^2 + 3ax + \sqrt{b} = 0$
(d) $\sqrt{b} x^2 + x - 3a = 0$

11. What is the sum of the squares of the roots of the equation $x^2 + 2x - 143 = 0$?

- (a) 170 (b) 180
(c) 190 (d) 290

12. If one of the roots of the equation $x^2 + ax - b = 0$ is, 1, then what is the value of $(a-b)$?

- (a) -1 (b) 1
(c) 2 (d) -2

13. If α and β are the roots of the equation. $x^2 - q(1+x) - r = 0$ then what is the value of $(1+\alpha)(1+\beta)$?
 (a) $1 - r$ (b) $q - r$
 (c) $1 + r$ (d) $q + r$
14. The solution of the simultaneous linear equations $2x + y = 6$ and $3y = 8 + 4x$ will also be satisfied by which one of the following linear equation?
 (a) $x + y = 5$ (b) $2x + y = 5$
 (c) $2x - 3y = 10$ (d) $2x + 3y = 6$
15. If the difference between the roots of $ax^2 + bx + c = 0$ is 1, then which one of the following is correct?
 (a) $b^2 = a(a+4c)$ (b) $a^2 = b(b+4c)$
 (c) $a^2 = c(a+4c)$ (d) $b^2 = a(b+4c)$
- Direction (for next two):** The equation formed by multiplying each root of $ax^2 + bx + c = 0$ by 2 is $x^2 + 36x + 24 = 0$.
16. What is the value of $b:c$?
 (a) 3:1 (b) 1:2
 (c) 1:3 (d) 3:2
17. Which one of the following correct?
 (a) $bc = a^2$ (b) $bc = 36a^2$
 (c) $bc = 72a^2$ (d) $bc = 108a^2$
18. The sum of the reciprocals of two alternate natural numbers is $\frac{7}{24}$. What is the sum of the numbers?
 (a) 12 (b) 13
 (c) 14 (d) 16
19. What is the value of $y = \sqrt{8+2\sqrt{8+2\sqrt{8+2\sqrt{8+\dots}}}}$
 (a) 10 (b) 8
 (c) 6 (d) 4
20. If α and β be the roots of the equation $(x-a)(x-b) = c$, $c \neq 0$. Then, the roots of the equation $(x-\alpha)(x-\beta) + c = 0$ are:
 (a) a, c (b) b, c
 (c) a, b (d) $a+b, a+c$
21. What is the sum of the roots of the equation $(2-\sqrt{3})x^2 - (7-4\sqrt{3})x + (2+\sqrt{3}) = 0$?
 (a) $2-\sqrt{3}$ (b) $2+\sqrt{3}$
 (c) $7-4\sqrt{3}$ (d) 4
22. If one root of the equation $ax^2 + bx + c = 0$, $a \neq 0$ is reciprocal of the other root, then which one of the following is correct?
 (a) $a = c$ (b) $b = c$
 (c) $a = -c$ (d) $b = 0$
23. If sum of squares of the roots of the equation $x^2 + kx - b = 0$ is $2b$, then what is k equal to?
 (a) 1 (b) b
 (c) $-b$ (d) 0
24. If 3 is the root of the equation $x^2 - 8x + k = 0$, then what is the value of k ?
 (a) -15 (b) 9
 (c) 15 (d) 24
25. What is the condition that one root of the equation $ax^2 + bx + c = 0$, $a \neq 0$ should be double the other?
 (a) $2a^2 = 9bc$ (b) $2b^2 = 9ac$
 (c) $2c^2 = 9ab$ (d) None of these
26. What is the solution set for the equation $x^4 - 26x^2 + 25 = 0$?
 (a) $\{-5, -1, 1, 5\}$ (b) $\{-5, -1\}$
 (c) $\{1, 5\}$ (d) $\{-5, 0, 1, 5\}$
27. If p, q, r are rational numbers, then the roots of the equation $x^2 - 2px + p^2 - q^2 + 2qr - r^2 = 0$ are:
 (a) Complex (b) Pure imaginary
 (c) Irrational (d) Rational
28. If $x + y \leq 4$, then the how many non-zero positive integer ordered pair (x, y) ?
 (a) 4 (b) 5
 (c) 6 (d) 8
29. What is the set of points (x, y) satisfying the equations $x^2 + y^2 = 4$ and $x + y = 2$?
 (a) $\{(2, 0), (-2, 0), (0, 2)\}$
 (b) $\{(0, -2), (0, 2)\}$
 (c) $\{(2, 0), (0, 2)\}$
 (d) $\{(2, 0), (-2, 0), (0, 2), (0, -2)\}$
30. If α and β are the roots of equation $4x^2 + 3x + 7 = 0$, then what is the value of $\alpha^{-2} + \beta^{-2}$?
 (a) $47/49$ (b) $49/47$
 (c) $-47/49$ (d) $-49/47$

ANSWER KEY

1.	(c)	2.	(a)	3.	(c)	4.	(c)	5.	(d)	6.	(c)	7.	(d)	8.	(b)	9.	(c)	10.	(a)
11.	(d)	12.	(a)	13.	(a)	14.	(a)	15.	(a)	16.	(a)	17.	(d)	18.	(c)	19.	(d)	20.	(c)
21.	(a)	22.	(a)	23.	(d)	24.	(c)	25.	(b)	26.	(a)	27.	(d)	28.	(c)	29.	(c)	30.	(c)

PRACTICE SHEET-4

- If the product of the roots of $x^2 - 3kx + 2k^2 - 1 = 0$ is 7 for a fixed k , then what is the nature of roots?
(a) Integral and positive
(b) Integral and negative
(c) Irrational
(d) Rational but not integral
- Which one of the following is the quadratic equation whose roots are reciprocal to the roots of the quadratic equation $2x^2 - 3x - 4 = 0$?
(a) $3x^2 - 2x - 4 = 0$ (b) $4x^2 + 3x - 2 = 0$
(c) $3x^2 - 4x - 2 = 0$ (d) $4x^2 - 2x - 3 = 0$
- The value of y which will satisfy the equations $2x^2 + 6x + 5y + 1$ and $2x + y + 3 = 0$ may be found by solving which one of the following equations?
(a) $y^2 + 14y - 7 = 0$ (b) $y^2 + 8y + 1 = 0$
(c) $y^2 + 10y - 7 = 0$ (d) $y^2 - 8y + 7 = 0$
- If a polynomial equation has rational coefficient and has exactly three real roots, then what is the degree of the polynomial?
(a) Equal to 3
(b) Greater than or equal to 3
(c) Strictly greater than 3
(d) Less than 3
- If α and β are the roots of $ax^2 + bx + c = 0$, then what is the value of $\left(\frac{1}{\alpha^2} - \frac{1}{\beta^2}\right)^2$?
(a) $\frac{b^2(b^2 - 4ac)}{c^4}$ (b) $\frac{b(b^2 - 4ac)}{c^2}$
(c) $\frac{(b^2 - 4ac)}{c^2}$ (d) $\frac{(b^2 - 4ac)}{c^4}$
- Which one of the following is one of the two consecutive positive integers, the sum of whose squares is 761?
(a) 15 (b) 20 (c) 24 (d) 25
- If $3^x + 27(3^{-x}) = 12$, then what is the value of x ?
(a) Only 1 (b) Only 2
(c) 1 or 2 (d) 0 or 1
- What is the magnitude of difference of the roots of $x^2 - ax + b = 0$?
(a) $\sqrt{a^2 - 4b}$ (b) $\sqrt{b^2 - 4a}$
(c) $2\sqrt{a^2 - 4b}$ (d) $\sqrt{b^2 - 4ab}$
- If $a + b = 2m^2$, $b + c = 6m$, $a + c = 2$, where m is a real number and $a \leq b \leq c$, then which one of the following is correct?
(a) $0 \leq m \leq 1/2$ (b) $-1 \leq m \leq 0$
(c) $1/3 \leq m \leq 1$ (d) $1 < m \leq 2$
- Students of a class are made to stand in rows. If one student is extra in a row, there would be two rows less. If one student is less in a row, there would be three rows more. Then, what is the number of students in the class?
(a) 65 (b) 55
(c) 60 (d) 50
- What is the ratio of sum of squares of roots to the product of the roots of the equation $7x^2 + 12x + 18 = 0$?
(a) 6 : 1 (b) 1 : 6
(c) -6 : 1 (d) -6 : 7
- If the roots of the equation $\frac{x(x-1)-(m+1)}{(x-1)(m-1)} = \frac{x}{m}$ are equal, then what is the value of m ?
(a) 1 (b) $1/2$
(c) 0 (d) $-1/2$

- 13.** What is the least integral value of k for which equation $x^2 - 2(k - 1)x + (2k + 1) = 0$ has both the roots positive?
 (a) 1 (b) $-1/2$
 (c) 4 (d) 0
- 14.** If one of the roots of the equation $ax^2 + x - 3 = 0$ is -1.5 , then what is the value of a ?
 (a) 4 (b) 3
 (c) 2 (d) -2
- 15.** r is a non-zero real number such that $r^{75} > r^{90}$. This is possible only when
 (a) $-1 < r < 0$ (b) $0 < r < 1$
 (c) $1 < r$ (d) $-1 < r < 1$
- 16.** When the roots of the quadratic equation $ax^2 + bx + c = 0$ are negative of reciprocals of each other, then which one of the following is correct?
 (a) $b = 0$ (b) $c = 0$
 (c) $a = c$ (d) $a = -c$
- 17.** What are the roots of the equation $\log_{10}(x^2 - 6x + 45) = 2$?
 (a) 9, -5 (b) -9 , 5
 (c) 11, -5 (d) -11 , 5
- 18.** The sum of the roots of the equation $\frac{1}{x+a} + \frac{1}{x+b} = \frac{1}{c}$ is zero. What is the product of the roots of the equation?
 (a) $-\frac{(a+b)}{2}$ (b) $\frac{(a+b)}{2}$
 (c) $-\frac{(a^2+b^2)}{2}$ (d) $\frac{(a^2+b^2)}{2}$
- 19.** For what value of k , will the roots of the equation $kx^2 - 5x + 6 = 0$ be in the ratio of 2 : 3?
 (a) 0 (b) 1
 (c) -1 (d) 2
- 20.** If α and β are the roots of the equation $x^2 + px + q = 0$, then $-\alpha^{-1}$, $-\beta^{-1}$ are the roots of which one of the following equations?
 (a) $qx^2 - px + 1 = 0$ (b) $q^2 + px + 1 = 0$
 (c) $x^2 + px - q = 0$ (d) $x^2 - px + q = 0$
- 21.** If one root of the equation $ax^2 + x - 3 = 0$ is -1 , then what is the other root?
 (a) $1/4$ (b) $1/2$
 (c) $3/4$ (d) 1
- 22.** If the equation $(a^2 + b^2)x^2 - 2(ac + bd)x + (c^2 + d^2) = 0$ has equal roots, then which one of the following is correct?
 (a) $ab = cd$ (b) $ad = bc$
 (c) $a^2 + c^2 = b^2 + d^2$ (d) $ac = bd$
- 23.** What is the solution of the equation

$$\sqrt{\frac{x}{x+3}} - \sqrt{\frac{x+3}{x}} = -\frac{3}{2}$$

 (a) 1 (b) 2
 (c) 4 (d) None of these
- 24.** What are the roots of the equation $4x - 3.2x^2 + 2 + 32 = 0$?
 (a) 1, 2 (b) 3, 4
 (c) 2, 3 (d) 1, 3
- 25.** If α and β are the roots of the equation $x^2 - x - 1 = 0$, then what is the value of $(\alpha^4 + \beta^4)$?
 (a) 7 (b) 0
 (c) 2 (d) None of these
- 26.** If sum as well as product of roots of a quadratic equation is 9, then what is the equation?
 (a) $x^2 + 9x - 18 = 0$ (b) $x^2 - 18x + 9 = 0$
 (c) $x^2 + 9x + 9 = 0$ (d) $x^2 - 9x + 9 = 0$
- 27.** If one root of the equation $2x^2 + 3x + c = 0$ is 0.5, then what is the value of c ?
 (a) -1 (b) -2
 (c) -3 (d) -4
- 28.** What is the condition that the equation $ax^2 + bx + c = 0$, where $a \neq 0$ has both the roots positive?
 (a) a , b and c are of same sign.
 (b) a and b are of same sign.
 (c) b and c have the same sign opposite to that of a .
 (d) a and c have the same sign opposite to that of b .
- 29.** The equation $(1 + n^2)x^2 + 2ncx + (c^2 - a^2) = 0$ will have equal roots, if
 (a) $c^2 = 1 + a^2$ (b) $c^2 = 1 - a^2$
 (c) $c^2 = 1 + n^2 + a^2$ (d) $c^2 = (1 + n^2)a^2$
- 30.** The equation whose roots are twice the roots of the equation $x^2 - 2x + 4 = 0$ is
 (a) $x^2 - 2x + 4 = 0$ (b) $x^2 - 2x + 16 = 0$
 (c) $x^2 - 4x + 8 = 0$ (d) $x^2 - 4x + 16 = 0$

ANSWER KEY

1.	(c)	2.	(b)	3.	(c)	4.	(a)	5.	(a)	6.	(b)	7.	(c)	8.	(a)	9.	(c)	10.	(c)
11.	(d)	12.	(d)	13.	(c)	14.	(c)	15.	(b)	16.	(d)	17.	(c)	18.	(c)	19.	(b)	20.	(a)
21.	(c)	22.	(b)	23.	(a)	24.	(c)	25.	(a)	26.	(d)	27.	(b)	28.	(d)	29.	(d)	30.	(d)

CDS PYQ

- X, Y and Z had taken a dinner together. The cost of the meal of Z was 20% more than that of Y and the cost of the meal of X was $\frac{5}{6}$ as much as the cost of the meal of Z. If Y paid Rs 100, then what was the total amount that all the three of them had paid?
[CDS 2013 (1)]
(a) Rs 285
(b) Rs 300
(c) Rs 355
(d) None of the above
- The system of equations $3x + y - 4 = 0$ and $6x + 2y - 8 = 0$ has:
[CDS 2013 (1)]
(a) A unique solution $x = 1, y = 1$
(b) A unique solution $x = 0, y = 4$
(c) No solution
(d) Infinite solution
- For what value of k, $(x + 5)$ is a factor of $6x^2 + kx + 10$?
[CDS 2013 (1)]
(a) 5
(b) 32
(c) 36
(d) 40
- A number consists of two digit whose sum is 10. If the digit of the number are reversed, then the number decreased by 36. Which of the following is/are correct?
I. The number is divisible by a composite number
II. The number is a multiple of a prime number
Select the correct answer using the codes given below:
[CDS 2013 (1)]
(a) Only I
(b) Only II
(c) Both I and II
(d) Neither I nor II
- Ten chairs and six tables together cost Rs 6200, three chairs and two tables together cost Rs 1900. The cost of 4 chairs and 5 tables is:
[CDS 2013 (1)]
(a) Rs 3000
(b) Rs 3300
(c) Rs 3500
(d) Rs 3800
- The difference in the root of the equation $2x^2 - 11x + 5 = 0$ is
[CDS 2013 (1)]
(a) 4.5
(b) 4
(c) 3.5
(d) 3
- If one of the roots of the equation $x^2 - bx + c = 0$ is the square of the other, then which of the following option is correct?
[CDS 2013 (1)]
(a) $b^3 = 3bc + c^2 + c$
(b) $c^3 = 3bc + b^2 + b$
(c) $3bc = c^3 + b^2 + b$
(d) $3bc = c^3 + b^3 + b^2$
- If $x + y - 7 = 0$ and $3x + y - 13 = 0$, then what is $4x^2 + y^2 + 4xy$ equal to?
[CDS 2013 (2)]
(a) 75
(b) 85
(c) 91
(d) 100
- A bus starts with some passengers. At the first stop, one-fifth of the passengers gets down and 40 passengers get in. At the second stop, half of the passengers gets down and 30 get in. The number of passengers now in 70. The number of passengers with which the bus started was:
[CDS 2013 (2)]

(a) 40
(c) 60

(b) 50
(d) 70

10. Consider the following statements in respect of the quadratic equation $ax^2 + bx + c = 0$, where $a \neq 0$.

I. The product of roots is equal to the sum of the roots

II. The roots of the equation are always unequal and real

Which of the statements given above is/are correct?

[CDS 2013 (2)]

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

11. If α and β are the roots of the equation $x^2 - x - 1 = 0$, then what is $\frac{\alpha^2 + \beta^2}{(\alpha^2 - \beta^2)(\alpha - \beta)}$ equal to?

[CDS 2013 (2)]

(a) $\frac{2}{5}$ (b) $\frac{3}{5}$
(c) $\frac{4}{5}$ (d) 4

12. Which one is one of the factors of:

$$x^2 + \frac{1}{x^2} + 8\left(x + \frac{1}{x}\right) + 14$$

[CDS 2013 (2)]

(a) $x + \frac{1}{x} + 1$ (b) $x + \frac{1}{x} + 3$
(c) $x + \frac{1}{x} + 6$ (d) $x + \frac{1}{x} + 7$

13. A positive number, when increased by 10 equals 200 times its reciprocal. What is number?

[CDS 2014 (1)]

(a) 100 (b) 10
(c) 20 (d) 200

14. The sum of two positive numbers x and y is 2.5 times their difference. If the product of numbers is 84, then what is the sum of those two numbers?

[CDS 2014 (1)]

(a) 26
(c) 22

(b) 24
(d) 20

15. Ravi's brother is 3 years elder to him. His father was 28 years of age when his sister was born while his mother was 26 years of age when he was born. If his sister was 4 years of age when his brother was born, the ages of Ravi's father and mother, respectively, when his brother was born were:

[CDS 2014 (1)]

(a) 32 years and 23 years
(b) 32 years and 29 years
(c) 35 years and 29 years
(d) 35 years and 33 years

16. Two chairs and one table cost Rs 700 and 1 chair and 2 tables costs Rs 800. If the cost m tables and m chairs is Rs 30,000, then what is m equal to?

[CDS 2014 (1)]

(a) 60 (b) 55
(c) 50 (d) 45

17. In solving a problem, one student makes a mistake in the coefficient of the first degree term and obtain -9 and -1 for the roots. Another student makes a mistake in the constant term of the equation and obtains 8 and 2 for the roots. The correct equation was:

[CDS 2014 (1)]

(a) $x^2 + 10x + 9 = 0$ (b) $x^2 - 10x + 16 = 0$
(c) $x^2 - 10x + 9 = 0$ (d) None of these

18. If m and n are the roots of the equation $ax^2 + bx + c = 0$, then the equation whose roots are $\frac{(m^2 + 1)}{m}$ and $\frac{(n^2 + 1)}{n}$ is:

[CDS 2014 (1)]

(a) $acx^2 + (ab + bc)x + b^2 + (a - c)^2 = 0$
(b) $acx^2 + (ab - bc)x + b^2 + (a - c)^2 = 0$
(c) $acx^2 - (ab - bc)x + b^2 - (a - c)^2 = 0$
(d) $acx^2 - (ab + bc)x + b^2 - (a - c)^2 = 0$

19. The value of $x^2 - 4x + 11$ can never be less than:

[CDS 2014 (1)]

(a) 7
(c) 11

(b) 8
(d) 22

- 20.** If the roots of the equation $(a^2 - bc)x^2 + 2(b^2 - ac)x + (c^2 - ab) = 0$ are equal, where $a, b, c \neq 0$, then which one of the following is correct?

[CDS 2014 (1)]

(a) $a+b+c=abc$ (b) $a^2+b^2+c^2=0$
(c) $a^3+b^3+c^3=0$ (d) $a^3+b^3+c^3=abc$

- 21.** If the roots of the equation $Ax^2 + Bx + C = 0$ are -1 and 1 , then which one of the following is correct?

[CDS 2014 (1)]

(a) A and C are both zero
(b) A and B are both positive
(c) A and C are both negative
(d) A and C are of opposite sign

- 22.** If m and n ($m > n$) are the roots of the equation $7(x + 2a)^2 + 3a^2 = 5a(7x + 23a)$, where $a > 0$, then what is $3m-n$ equal to?

[CDS 2014 (2)]

(a) $12a$ (b) $14a$
(c) $15a$ (d) $18a$

- 23.** If one the roots of the equation $px^2 + qx + r = 0$ is three times the other, then which one of the following relations is correct?

[CDS 2014 (2)]

(a) $3q^2 = 16pr$ (b) $a^2 = 24pr$
(c) $p = q + r$ (d) $p + q + r = 1$

- 24.** If m and n are the roots of the equation $x^2 + ax + b = 0$ and m^2, n^2 are the roots of the equation $x^2 - cx + d = 0$, then which of the following is/are correct?

1. $2b - a^2 = c$
2. $b^2 = d$

Select the correct answer using the codes given below:

[CDS 2014 (2)]

(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

- 25.** If $(x+k)$ is the common factor of $x^2 + ax + b$ and $x^2 + cx + d$ of and then what is k equal to?

[CDS 2014 (2)]

(a) $(d-b)/(c-a)$
(c) $(d+b)/(c+a)$

(b) $(d-b)/(a-c)$
(d) $(d-b)/(c+a)$

- 26.** If the equation $x^2 + 2(1+k)x + k^2 = 0$ has equal roots, then what is the value of k ?

[CDS 2014 (2)]

(a) $1/2$ (b) $-1/2$
(c) 1 (d) -1

- 27.** Consider the following statements:

1.The equation $1990x - 173y = 11$ has no solution in integers for x and y .
2.The equation $3x - 12y = 7$ has no solution in integers for x and y .

Which of the above statements is/are correct?

[CDS 2015 (1)]

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

- 28.** For which value of k does the pair of equations $x^2 - y^2 = 0$ and $(x - k)^2 + y^2 = 1$ yield a unique positive solution of x ?

[CDS 2015 (1)]

(a) 2 (b) 0
(c) $\sqrt{2}$ (d) $-\sqrt{2}$

- 29.** The sign of the quadratic polynomial $ax^2 + bx + c$ is always positive if:

[CDS 2015 (1)]

(a) a is positive and $b^2 - 4ac \leq 0$
(b) a is positive and $b^2 - 4ac \geq 0$
(c) a can be any real number and $b^2 - 4ac \leq 0$
(d) a can be any real number and $b^2 - 4ac \geq 0$

- 30.** If $k = x - y + 2z$ where $-2 \leq x \leq 1$, $-1 \leq y \leq 2$ and $3 \leq z \leq 6$, then which one of the following is correct?

[CDS 2015 (2)]

(a) $0 \leq k \leq 9$ (b) $5 \leq k \leq 11$
(c) $2 \leq k \leq 14$ (d) $2 \leq k \leq 11$

- 31.** The number of pairs (x, y) where x, y are integers satisfying the equation $21x + 48y = 5$ is:

[CDS 2015 (2)]

(a) Zero (b) One
(c) Two (d) Infinity

- 32.** Let x and y be positive integers such that x is prime and y is composite. Which of the following statements are correct?

1. $(y-x)$ can be an even integer
 2. xy can be an even integer
 3. $0.5(x+y)$ can be an even integer
- Select the correct answer using the code given below:

[CDS 2015 (2)]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

33. If the roots of the quadratic equations $x^2 - 4 - \log_{10} N = 0$ are all real, then the minimum value of N is:

[CDS 2015 (2)]

- (a) $\frac{1}{100}$ (b) $\frac{1}{1000}$
(c) $\frac{1}{10000}$ (d) 10000

34. The difference of maximum values of the expressions $(6 + 5x - x^2)$ and $(y - 6 - y^2)$ for any real values of x and y is:

[CDS 2015 (2)]

- (a) 16 (b) 17
(c) 18 (d) 19

35. What is $\sqrt{4 + \sqrt{4 - \sqrt{4 + \sqrt{4 - \dots}}}}$ equal to?

[CDS 2015 (2)]

- (a) 3 (b) $\frac{\sqrt{13}-1}{2}$
(c) $\frac{\sqrt{13}+1}{2}$ (d) 0

36. A tin of oil was $\frac{4}{5}$ full. When 6 bottles of oil were taken out from this tin and 4 bottles of oil were poured into it, it was $\frac{3}{4}$ full. Oil of how many bottles can the tin contain? (All bottles are of equal volume)

[CDS 2015 (2)]

- (a) 35 (b) 40
(c) 45 (d) 50

37. If the equations $x^2 - px + q = 0$ and $x^2 + qx - p = 0$ have a common root, then which one of the following is correct?

[CDS 2016 (1)]

- (a) $p-q=0$ (b) $p+q-2=0$
(c) $p+q-1=0$ (d) $p-q-1=0$

38. The value of k , for which the system of equation $3x - ky - 20 = 0$ and $6x - 10y + 40 = 0$ has no solution, is:

[CDS 2016 (1)]

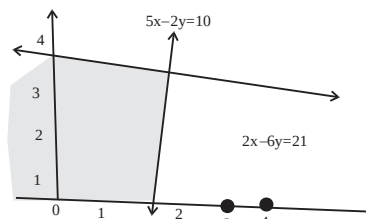
- (a) 10 (b) 6
(c) 5 (d) 3

39. There are three brothers. The sums of ages of two of them at a time are 4 years, 6 years and 8 years. The age difference between the eldest and the youngest is:

[CDS 2016 (1)]

- (a) 3years (b) 4years
(c) 5 years (d) 6 years

40.



The linear in equations, for which the shaded area in the figure given above is the solution set, are:

[CDS 2016 (1)]

- (a) $2x+6y \leq 21, 5x-2y \leq 10$
(b) $2x+6y \leq 21, 5x-2y \geq 10$
(c) $2x+6y \geq 21, 5x-2y \leq 10$
(d) $2x+6y \geq 21, 5x-2y \geq 10$

41. If a and b are negative real numbers and c is a positive real number, then which of the following is/are correct?

1. $a - b < a - c$
2. If $a < b$ then $\frac{a}{c} < \frac{b}{c}$
3. $\frac{1}{b} < \frac{1}{c}$

Select the correct answer using the code given below:

[CDS 2016 (1)]

- (a) 1 (b) 2 only
(c) 3 only (d) 2 and 3

42. If the roots of the equation $lx^2 + mx + m = 0$ are in the ratio $p : q$, then:

$\sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} + \sqrt{\frac{m}{l}}$ is equal to:

[CDS 2016 (1)]

- (a) 0 (b) 1
(c) 2 (d) 3

43. If $\sqrt{3x^2 - 7x - 30} - \sqrt{2x^2 - 7x - 5} = x - 5$ has α and β as its roots, then the value of $\alpha\beta$ is:

[CDS 2016 (1)]

- (a) -15 (b) -5
(c) 0 (d) 5

44. Let p and q be non-zero integers. Consider the polynomial $A(x) = x^2 + px + q$. It is given that $(x - m)$ and $(x - km)$ are linear factors of $A(x)$, where m is a non-zero integer and k is a positive integer, $k \geq 2$. Which one of the following is correct?

[CDS 2016 (1)]

- (a) $(k+1)^2 p^2 = kg$ (b) $(k+1)^2 q = kp^2$
(c) $k^2 q = (k+1) p^2$ (d) $k^2 p^2 = (k+1)^2 q$

45. If the linear factors of $ax^2 - (a^2 + 1)x + a$ are p and q then $p + q$ is equal to:

[CDS 2016 (1)]

- (a) $(x - 1)(a + 1)$ (b) $(x + 1)(a + 1)$
(c) $(x - 1)(a - 1)$ (d) $(x + 1)(a - 1)$

46. If the sum of the roots of $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals, then which one of the following relations is correct?

[CDS 2016 (1)]

- (a) $ab^2 + bc^2 = 2a^2 c$ (b) $ac^2 + bc^2 = 2b^2 a$
(c) $ab^2 + bc^2 = a^2 c$ (d) $a^2 + b^2 + c^2 = 1$

47. Under what condition on p and q , one of the roots of the equation $x^2 + px + q = 0$ is the square of the other?

[CDS 2016 (1)]

- (a) $1 + q + q^2 = 3pq$ (b) $1 + q + p^2 = 3pq$
(c) $p^3 + q + q^2 = 3pq$ (d) $q^3 + p + p^2 = 3pq$

48. The solution of the in-equation:

$$1 + \frac{1}{x} - \frac{1}{x^2} \geq 0 \text{ is (given that } x \neq 0)$$

[CDS 2016 (1)]

- (a) $x > 0$
(b) $x < 0$
(c) $\frac{-1-\sqrt{5}}{2} \leq x \leq \frac{-1+\sqrt{5}}{2}$
(d) $x \leq \frac{-1-\sqrt{5}}{2}$ or $x \geq \frac{-1+\sqrt{5}}{2}$

49. If $4 \times 2^y = 128$ and $3^3 \times 3^{2y} - 9^{xy} = 0$, then the value of $x + y$ can be equal to:

[CDS 2016 (1)]

- (a) 7 (b) 5
(c) 3 (d) 1

50. Which of the points $P(5, -1)$, $Q(3, -2)$ and $R(1, 1)$ lie in the solution of the system of in-equations $x + y \leq 4$ and $x - y \geq 2$?

[CDS 2016 (1)]

- (a) Q and R only (b) P and R only
(c) P and Q only (d) P , Q and R

51. If λ is an integer and α, β are the roots of $4x^2 - 16x + \frac{\lambda}{4} = 0$ such that $1 < \alpha < 2$ and $2 < \beta < 3$, then how many values can λ take?

[CDS 2016 (2)]

- (a) 3 (b) 9
(c) 14 (d) 15

52. If α and β are the two zeros of the polynomial $25x^2 - 15x + 2$, then what is a quadratic polynomial whose zeros are $(2\alpha)^{-1}$ and $(2\beta)^{-1}$?

[CDS 2016 (2)]

- (a) $x^2 + 30x + 2$ (b) $8x^2 - 30x + 25$
(c) $8x^2 - 30x$ (d) $x^2 + 30x$

53. If p and q are the roots of $x^2 + px + q = 0$, then which of the following is correct?

[CDS 2016 (2)]

- (a) $p=0$ or 1 (b) $p = 1$ only
(c) $p = -2$ or 0 (d) $p = -2$ only

54. The pair of linear equations $kx + 3y + 1 = 0$ and $2x + y + 3 = 0$ intersect each other, if:

[CDS 2017 (1)]

- (a) $k=6$ (b) $k \neq 6$
(c) $k=0$ (d) $k \neq 0$

55. The system of equations $2x + 4y = 6$ and $4x + 8y = 8$ is:

[CDS 2017 (1)]

- (a) Consistent with a unique solution
(b) Consistent with infinitely many solution
(c) Inconsistent
(d) None of the above

56. If α and β are the roots of the quadratic equation $2x^2 + 6x + k = 0$, where $k < 0$, then what is the maximum value of $\left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right)$?

[CDS 2017 (1)]

- (a) 2 (b) -2
(c) 9 (d) -9

57. If one root of $(a^2 - 5a + 3)x^2 + (3a - 1)x + 2 = 0$ is twice the other, then what is the value of 'a'?

[CDS 2017 (1)]

(a) $\frac{2}{3}$

(b) $-\frac{2}{3}$

(c) $\frac{1}{3}$

(d) $-\frac{1}{3}$

58. $(x+4)$ is a factor of which one of the following expressions?

[CDS 2017 (1)]

(a) $x^2 - 7x + 44$

(b) $x^2 - 7x - 44$

(c) $x^2 - 7x - 44$

(d) $x^2 + 7x + 44$

59. If α and β are the roots of the equation $x^2 + px + q = 0$, then what is $\alpha^2 + \beta^2$ equal to?

[CDS 2017 (1)]

(a) $p^2 - 2q$

(b) $a^2 - 2p$

(c) $p^2 + 2q$

(d) $q^2 - q$

60. Aman and Alok attempted to solve a quadratic equation. Aman made a mistake in writing down the constant term and ended up in roots (4,3). Alok made a mistake in writing down the coefficient of x to get roots (3,2). The correct roots of the equation are:

[CDS 2017 (1)]

(a) -4, -3

(b) 6, 1

(c) 4, 3

(d) -1, -1

61. What number must be subtracted from both the numerator and the denominator of the fraction $\frac{27}{35}$ so that it becomes $\frac{2}{3}$?

[CDS 2017 (1)]

(a) 6

(b) 8

(c) 9

(d) 11

62. The value of x which satisfy the equation $5^{1+x} + 5^{1-x} = 26$ are:

[CDS 2017 (1)]

(a) -1, 1

(b) 0, 1

(c) 1, 2

(d) -1, 0

63. Sunil wants to spend Rs200 on two types of sweets, costing Rs 7 and Rs 10 respectively. What is the maximum number of sweet he can get so that no money is left over?

[CDS 2017 (1)]

(a) 25

(b) 26

(c) 27

(d) 28

64. What is the value of u in the system of equations $3(2u + v) = 7uv$, $3(u + 3v) = 11uv$?

[CDS 2017 (2)]

(a) 0

(b) $\frac{1}{4}$

(c) $\frac{1}{2}$

(d) 1

65. What is the positive value of m for which the roots of the equation $12x^2 + mx + 5 = 0$ are in the ratio 3:2?

[CDS 2017 (2)]

(a) $5\sqrt{10}$

(b) $\frac{5\sqrt{10}}{12}$

(c) $\frac{5}{12}$

(d) $\frac{12}{5}$

66. If the roots of the equation $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ are equal, then which one of the following is correct?

[CDS 2017 (2)]

(a) $2b = a + c$

(b) $b^2 = ac$

(c) $\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$

(d) $\frac{1}{b} = \frac{1}{a} + \frac{1}{c}$

67. If k is an integer, then $x^2 + 7x - 14 \left(k^2 - \frac{7}{8} \right) = 0$ has:

[CDS 2017 (2)]

(a) Both integral roots

(b) At least one integral root

(c) No integral root

(d) Both positive integral roots

68. What is the value of α ($\alpha \neq 0$) for which $x^2 - 5x + \alpha$, and $x^2 - 7x + 2\alpha$ have a common factor?

[CDS 2017 (2)]

(a) 6

(b) 4

(c) 3

(d) 2

69. The value of $\sqrt{1 + \sqrt{1 + \sqrt{1 + \dots}}}$:

[CDS 2017 (2)]

(a) Equals to 1

(b) Lies between 0 and 1

(c) Lies between 1 and 2

(d) Is greater than 2

70. If $65x - 33y = 97$ and $33x - 65y = 1$, then what is xy equal to?

[CDS 2018 (1)]

(a) 2

(b) 3

(c) -2

(d) -3

71. If the roots of the equation $px^2 + x + r = 0$ are reciprocal to each other, then which one of the following is correct?

[CDS 2018 (1)]

(a) $p=2r$

(b) $p=r$

(c) $2p=r$

(d) $p=4r$

72. If α and β are the roots of the equation $ax^2 + bx + c = 0$, then what is the value of expression $(\alpha+1)(\beta+1)$?

[CDS 2018 (1)]

- (a) $\frac{a+b+c}{a}$ (b) $\frac{b+c-a}{a}$
(c) $\frac{a-b+c}{a}$ (d) $\frac{a+b-c}{a}$

73. A quadratic polynomial $ax^2 + bx + c$ is such that when it is divided by x , $(x-1)$ and $(x+1)$, the remainders are 3, 6 and 4 respectively. What is the value of $(a+b)$?

[CDS 2018 (1)]

- (a) 3 (b) 2
(c) 1 (d) -1

74. $\frac{1}{25}$ of the students who registered did not appear for the examination, $\frac{11}{20}$ of those who appeared passed. If the number of registered students is 2000, the number who passed is:

[CDS 2018 (1)]

- (a) 1920 (b) 1056
(c) 1020 (d) 864

75. The age of a woman is a two-digit integer. On reversing this integer, the new integer is the age of her husband who is elder to her. The difference between their ages is one-eleventh of their sum. What is the difference between their ages?

[CDS 2018 (1)]

- (a) 8 yrs (b) 9 yrs
(c) 10 yrs (d) 11 yrs

76. If $x^2 - 6x - 27 > 0$, then which one of the following is correct?

[CDS 2018 (2)]

- (a) $-3 < x < 9$ (b) $x < 9$ or $x > -3$
(c) $x > 9$ or $x < -3$ (d) $x < -3$

77. If the sum of the square of three consecutive natural numbers is 110, then the sum of their cubes is:

[CDS 2018 (2)]

- (a) 625 (b) 654
(c) 684 (d) 725

78. If α and β are the roots of the equation $ax^2 + bx + c = 0$, then the value of $\frac{1}{a\alpha+b} + \frac{1}{a\beta+b}$ is:

[CDS 2018 (2)]

- (a) $\frac{a}{bc}$ (b) $\frac{b}{ac}$
(c) $\frac{c}{ab}$ (d) $\frac{1}{abc}$

79. The minimum value of the expression $2x^2 + 5x + 5$ is:

[CDS 2018 (2)]

- (a) 5 (b) $15/8$
(c) $-15/8$ (d) 0

80. According to Mr. Sharma's will, half of his property goes to his wife and the rest is equally divided between his two sons, Ravi and Raj. Some years later, Ravi dies and leaves half of his property to his widow and rest to his brother Raj. When Raj dies he leaves half of his property to his widow and remaining to his mother, who is still alive. The mother now owns Rs 88,000 worth of the property. The total worth of the property of Mr. Sharma was :

[CDS 2018 (2)]

- (a) Rs 1,00,000 (b) Rs 1,24,000
(c) Rs 1,28,000 (d) Rs 1,32,000

81. X bought 4 bottles of lemon juice and Y bought one bottle of orange juice. Orange juice per bottle costs twice the cost of lemon juice per bottle. Z bought nothing but contributed Rs 50 for his share of the drink which they mixed together and shared the cost equally. If Z's Rs 50 is covered from his share, then what is the cost of one bottle of orange juice?

[CDS 2018 (2)]

- (a) Rs 75 (b) Rs 50
(c) Rs 46 (d) Rs 30

82. The solution of linear inequalities $x + y \geq 5$ and $x - y \leq 3$ lies:

[CDS 2018 (2)]

- (a) Only in the first quadrant
(b) In the first and second quadrants
(c) In the second and third quadrants
(d) In the third and fourth quadrants

83. It is given that the equations $x^2 - y^2 = 0$ and $(x-a)^2 + y^2 = 1$ have single positive solution. For this, the value of 'a' is:

[CDS 2018 (2)]

- (a) $\sqrt{2}$ (b) 2
(c) $-\sqrt{2}$ (d) 1

- 84.** Which of the following pair of numbers is the solution of the equation $3^{x+2} + 3^{-x} = 10$?
[CDS 2019 (1)]
(a) 0, 2 (b) 0, -2
(c) 1, -1 (d) 1, 2
- 85.** The inequality $3^N > N^3$ holds when:
[CDS 2019 (1)]
(a) N is any natural number
(b) N is a natural number greater than 2
(c) N is a natural number greater than 3
(d) N is a natural number except 3
- 86.** A man who recently died left a sum of Rs 3,90,000 to be divided among his wife, five sons and four daughters. He directed that each son should receive 3 times as much as each daughter receives and that each daughter should receive twice as much as their mother receives. What was the wife's share?
[CDS 2019 (1)]
(a) Rs 14,000 (b) Rs 12,000
(c) Rs 10,000 (d) Rs 9,000
- 87.** If p and q are the roots of the equation $x^2 - 15x + r = 0$ and $p - q = 1$, then what is the value of r?
[CDS 2019 (1)]
(a) 55 (b) 56
(c) 60 (d) 64
- 88.** For the in-equality $x^2 - 7x + 12 > 0$, which one of the following is correct?
[CDS 2019 (1)]
(a) $3 < x < 4$
(b) $-\infty < x < 3$ only
(c) $4 < x < \infty$ only
(d) $-\infty < x < 3$ or $4 < x < \infty$
- 89.** If the sum of a real number and its reciprocal is $\frac{26}{5}$, then how many such numbers are possible?
[CDS 2019 (2)]
(a) None (b) one
(c) Two (d) Four
- 90.** If the equations $x^2 + 5x + 6 = 0$ and $x^2 + kx + 1 = 0$ have a common root, then what is the value of k?
[CDS 2019 (2)]
(a) $-\frac{5}{2}$ or $-\frac{10}{3}$ (b) $\frac{5}{2}$ or $\frac{10}{3}$
(c) $\frac{5}{2}$ or $-\frac{10}{3}$ (d) $-\frac{5}{2}$ or $\frac{10}{3}$
- 91.** What is $(x - a)(x - b)(x - c)$ equal to?
[CDS 2019 (2)]
(a) $x^3 - (a + b + c)x^2 + (bc + ca + ab)x - abc$
(b) $x^3 - (a + b + c)x^2 + (bc + ca + ab)x + abc$
(c) $x^3 - (bc + ca + ab)x^2 + (a + b + c)x - abc$
(d) $x^3 + (bc + ca + ab)x^2 - (a + b + c)x - abc$
- 92.** A person carries Rs500 and wants to buy apples and oranges out of it. If the cost of one apple is Rs5 and the cost of one orange is Rs7, then what is the number of ways in which a person can buy both apples and oranges using total amount?
[CDS 2019 (2)]
(a) 10 (b) 14
(c) 15 (d) 17
- 93.** A real number x is such that $(x - x^2)$ is maximum. What is x equal to?
[CDS 2019 (2)]
(a) -1.5 (b) -0.5
(c) 0.5 (d) 1.5
- 94.** The sum of the squares of four consecutive natural numbers is 294. What is the sum of the numbers?
[CDS 2019 (2)]
(a) 38 (b) 34
(c) 30 (d) 26
- 95.** The equation $x^2 + px + q = 0$ has roots equal to p and q where $q \neq 0$. What are the values of p and q respectively?
[CDS 2019 (2)]
(a) 1, -2 (b) 1, 2
(c) -1, 2 (d) -1, -2
- 96.** If (b-6) is one root of the quadratic equation $x^2 - 6x + b = 0$, where b is an integer, then what is the maximum value of b^2 ?
[CDS 2019 (2)]
(a) 36 (b) 49
(c) 64 (d) 81
- 97.** What is the maximum value of the expression $\frac{1}{x^2 + 5x + 10}$?
[CDS 2019 (2)]
(a) $\frac{15}{4}$ (b) $\frac{15}{2}$
(c) 1 (d) $\frac{4}{15}$
- 98.** Two numbers p and q are such that the quadratic equation $px^2 + 3x + 2q = 0$ has -6

as the sum and the product of the roots.
What is the value of $(p-q)$?

[CDS 2019 (2)]

- (a) -1 (b) 1
(c) 2 (d) 3

99. If α and β are the roots of the quadratic equation $x^2 + kx - 15 = 0$ such that $\alpha - \beta = 8$, then what is the positive value of k ?

[CDS 2020 (1)]

- (a) 2 (b) 3
(c) 4 (d) 5

100. Students of a class are made to sit in rows of equal number of chairs. If number of students is increased by 2 in each row, then the number of rows decreases by 3. If number of students is increased by 4 in each row, then the number of rows decreases by 5. What is the number of students in the class?

[CDS 2020 (1)]

- (a) 100 (b) 105
(c) 110 (d) 120

101. How many integral values of x and y satisfy the equation $5x + 9y = 7$, where $-500 < x < 500$ and $-500 < y < 500$?

[CDS 2020 (1)]

- (a) 110 (b) 111
(c) 112 (d) None

102. If $x^2 + 9y^2 = 6xy$, then what is $y : x$ equal to?

[CDS 2020 (1)]

- (a) 1 : 3 (b) 1 : 2
(c) 2 : 1 (d) 3 : 1

103. The number of different solutions of the equations $x + y + z = 12$, where each of x , y and z is a positive integer, is:

[CDS 2020 (2)]

- (a) 53 (b) 54
(c) 55 (d) 56

104. A 60 page book has n lines per page. If the number of lines were reduced by 3 in each page, the number of pages would have to be increased by 10 to give the same writing space. What is the value of n ?

[CDS 2020 (2)]

- (a) 18 (b) 21
(c) 24 (d) 30

105. If the equation $4x^2 - 2kx + 3k = 0$ has equal roots, then what are the values of k ?

[CDS 2021 (1)]

- (a) 4, 12 (b) 4, 8
(c) 0, 12 (d) 0, 8

106. If the sum as well as the product of the roots of the equation $px^2 - 6x + q = 0$ is 6, then what is $(p+q)$ equal to?

[CDS 2021 (1)]

- (a) 8 (b) 7
(c) 6 (d) 5

107. Which one of the following fractions will have minimum change in its value if 3 is added to both the numerator and the denominator of all the fractions?

[CDS 2021 (1)]

- (a) $\frac{2}{3}$ (b) $\frac{3}{4}$
(c) $\frac{4}{5}$ (d) $\frac{5}{6}$

108. The sum of the reciprocals of two alternate natural numbers is $\frac{7}{24}$. What is the sum of the numbers?

[CDS 2021 (1)]

- (a) 12 (b) 13
(c) 14 (d) 16

109. Which one of the following equations does not have real roots?

[CDS 2021 (2)]

- (a) $2x^2 + 16x + 3 = 0$ (b) $2x^2 + 10x - 1 = 0$
(c) $x^2 - 8x + 1 = 0$ (d) $4x^2 + 9x + 6 = 0$

110. The sum and the product of the roots of a quadratic equation are 7 and 12 respectively. If the bigger root is halved and the smaller root is doubled, then what is the resulting quadratic equation?

[CDS 2021 (2)]

- (a) $x^2 - 6x + 12 = 0$ (b) $x^2 - 8x + 12 = 0$
(c) $x^2 + 8x + 12 = 0$ (d) $x^2 - 10x + 12 = 0$

111. For which values of k , does the equation $x^2 - kx + 2 = 0$ have real and distinct solutions?

[CDS 2021 (2)]

- (a) $-2\sqrt{2} < k < 2\sqrt{2}$ (b) $k < -2\sqrt{2}$ only
(c) $k > 2\sqrt{2}$ only (d) $k < -2\sqrt{2}$ or $k > 2\sqrt{2}$

112. 5 pencils, 6 notebooks and 7 erasers Cost Rs250; whereas 6 pencils, 4 notebooks and 2 erasers cost Rs180. What is the cost of 2 notebooks and 4 erasers?

[CDS 2021 (2)]

- (a) Rs 90 (b) Rs 75
(c) Rs 60 (d) Rs 40

113. If p and q ($p > q$) are the roots of the equation $x^2 - 60x + 899 = 0$, then which one of the following is correct?

[CDS 2021 (2)]

- (a) $p - q - 1 = 0$ (b) $p - 2q + 27 = 0$
(c) $2p - q - 30 = 0$ (d) $3p - 2q - 43 = 0$

114. If one of the roots of the equation $ax^2 - 4ax + 15 = 0$ is $\frac{3}{2}$, then what is the sum of the squares of the roots?

[CDS 2021 (2)]

- (a) $\frac{15}{2}$ (b) $\frac{17}{2}$
(c) $\frac{19}{2}$ (d) $\frac{21}{2}$

115. The present age of a father is equal to sum of the ages of his 4 children. After ten years the sum of the ages of the children will be 1.6 times the age of their father. What is the present age of father?

[CDS 2021 (2)]

- (a) 36 years (b) 40 years
(c) 42 years (d) 45 years

116. The sum of numerator and denominator of a fraction is 10. If the numerator is increased by 3 and denominator is decreased by 1, the fraction becomes 1. What is the difference between numerator and denominator of the fraction?

[CDS 2021 (2)]

- (a) 2 (b) 3
(c) 4 (d) 5

117. If the system of equation $7x + ky = 27$ and $kx + 7y = 19$ have unique solution, then which one of the following is correct?

[CDS 2021 (2)]

- (a) $k \neq 7$ (b) $k \neq 13$
(c) $k = 7$ (d) $k = 13$

118. A person agrees to work for 30 days, on a condition for every day's work he should receive Rs500, and that for every day's absence from work he should forfeit Rs100. At the end of the time he received Rs 11,400. How many days did he work?

[CDS 2021 (2)]

- (a) 20 (b) 21
(c) 24 (d) 25

119. Consider a question and two statements:
Question:

Is $3x + 2y$ positive

Statement-I: $x^3 = -29.8$

Statement-II: $y^3 = 3x$

Which one of the following is correct in respect of the question and the statements?

[CDS-2022-1]

- (a) Statement-I alone is sufficient to answer the question
(b) Statement-II alone sufficient to answer the question
(c) Both Statements-I & II are together sufficient to answer the question.
(d) Both Statement-I & II are not sufficient to answer the question.

120. Consider a question and two statements:

Question:

Does the equation $ax^2 + bx + c^2 = 0$

Have real roots of opposite sign

Statement-I: The discriminates $D > 0$

Statement-II: $c/a > 0$

Which one of the following is correct in respect of the question and the statements?

[CDS-2022-1]

- (a) Statement-I alone is sufficient to answer the question
(b) Statement-II alone sufficient to answer the question
(c) Both Statements-I & II are together sufficient to answer the question.
(d) Both Statement-I & II are not sufficient to answer the question.

121. How many quadratic equations have the sum of their roots equal to the product of their roots?

[CDS-2022-1]

- (a) Zero (b) One
(c) Two (d) Infinitely many

122. If α and β are the roots of the quadratic equation $x^2 + \alpha x + \beta = 0$, where $\beta \neq 0$, then what is the value of $\alpha - \beta$?

[CDS-2022-1]

- (a) 4 (b) 3
(c) -1 (d) -3

123. If the sum of the roots of the equation $x^2 - k^2x + 30kx - 161x - 64 = 0$ is zero, then what is the difference of the roots?

[CDS-2022-1]

- (a) 15 (b) 16
(c) 17 (d) 18

124. A piece of cloth costs ₹10,000. If a 2 m longer piece of the same cloth is purchased for the same amount, it would cost ₹250

less per meter. What is the original length of the piece of cloth?

[CDS-2022-1]

- (a) 8m (b) 10m
(c) 12m (d) 16m

125. What is the condition that the roots of the equation $ax^2+bx+c=0$ are in the ratio $c : 1$?

[CDS-2022-1]

- (a) $b^2 = a(c+1)^2$ (b) $a^2 = bc(c+1)^2$
(c) $b^2 = a(c-1)^2$ (d) $ab^2 = (c+1)^2$

126. Two sides of a triangle forming a right angle are $6x^2$ and $(2x^2 - 1)$. If the area of the triangle is 84 square units, then what is the perimeter of the triangle?

[CDS-2022-1]

- (a) 51 units (b) 53 units
(c) 56 units (d) 59 units

127. If squaring a positive real number x is same as adding 12, then what is x equal to?

[CDS-2022-1]

- (a) 2 (b) 3
(c) 4 (d) 5

128. How many minutes are there in x weeks and x days?

[CDS-2022-1]

- (a) $11520x$ (b) $5760x$
(c) $480x$ (d) $192x$

129. What are the values of k for which the polynomial $(k-3)x^2 - kx - 1$ has no real linear factors?

[CDS 2022 (II)]

- (a) $k < -6$ (b) $-6 < k < 2$
(c) $2 < k < 6$ (d) $k > 6$

130. The value of a 2-digit number is 5 times the sum of the digits. What is the product of the digits?

[CDS 2022 (II)]

- (a) 15 (b) 18
(c) 20 (d) 27

131. If the sum of the squares of the roots of the equation $x^2 - 14x + k = 0$ is 100, then what is the value of k ?

[CDS 2022 (II)]

- (a) 42 (b) 48
(c) 52 (d) 56

132. Consider the following statements in respect of the polynomial $a(b-c)(x-b)(x-c) + b(c-a)(x-c)(x-a) + c(a-b)(x-a)(x-b)$:

1. the coefficient of x^2 is 0

2. The coefficient of x is $(a-b)(b-c)(c-a)$
Which of the statements given above is/are correct?

[CDS - 2023 (1)]

- (a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

133. For what values of m , is $mx^2 + mx + 8x + 9$ a perfect square?

[CDS - 2023 (1)]

- (a) 1, 4 (b) 4, 9
(c) 9, 16 (d) 4, 16

134. If α and β are the roots of the equation $x^2 - 7x + 1 = 0$, then what is the value of $\alpha^4 + \beta^4$?

[CDS - 2023 (1)]

- (a) 2207 (b) 2247
(c) 2317 (d) 2337

135. The age of Q exceeds the age of P by 3 years. The age of R is twice the age of S. Further, the age difference of R and S is 30 years. What is the sum of the ages of P and Q?

[CDS - 2023 (1)]

- (a) 35 years (b) 38 years
(c) 39 years (d) 45 years

136. If a, b, c are non-zero real numbers such that $a + b + c = 0$, then what are the roots of the equation $ax^2 + bx + c = 0$?

[CDS - 2023 (1)]

- (a) 2, $1 + (c/a)$ (b) 1, a/c
(c) 1, c/a (d) 2, $(c/a)-1$

137. If the roots of the equation $x^2 - bx + c = 5$ differ by 5, then which one of the following is correct?

[CDS - 2023 (1)]

- (a) $b^2 = 4c + 5$ (b) $c^2 = 4b - 5$
(c) $b^2 + c^2 = 5$ (d) $b^2 - c^2 = 5$

138. The sum of digits of a 2-digit number is 12. When the digits are reversed, the number becomes greater by eighteen. What is the difference between the digits in the number?

[CDS - 2023 (1)]

- (a) 1 (b) 2
(c) 3 (d) 4

139. Question: What are the unique values of a, b and c if 2 is a root of the equation $ax^2 + bx + c = 0$?

Statement I: Ratio of c to a is 1

Statement II: Ratio of b to a is $(-5/2)$

[CDS - 2023 (1)]

(a) choose this option if the question can be answered by one of the statements alone but not by the other.

(b) choose this option if the question can be answered by either statement alone.

(c) Choose this option if the question can be answered by using both the statements together, but cannot be answered by using either statement alone

(d) Choose this option if the question cannot be answered even by using both statements together

- 140.** Let α and β be the roots of the equation $\frac{1}{x+a+b} = \frac{1}{x} + \frac{1}{a} + \frac{1}{b}$; $a \neq 0$, $b \neq 0$, $x \neq 0$. Which one of the following is a quadratic equation whose roots are α^2 and β^2 ?

[CDS-2023 (2)]

- (a) $x^2 + (a^2 + b^2)x + a^2b^2 = 0$
 (b) $x^2 - (a^2 + b^2)x + a^2b^2 = 0$
 (c) $x^2 - (a^2 + b^2)x - a^2b^2 = 0$
 (d) $x^2 + (a^2 + b^2)x - a^2b^2 = 0$

- 141.** If $x = \frac{6}{7 - \frac{6}{7 - \frac{6}{7 - \frac{6}{7 - x}}}}$ the value of $x^2 - 3x + 2$

equal to?

[CDS-2023 (2)]

- (a) 0 (b) 1
 (c) 18 (d) 20

Consider the following the next (05) items that follow:

A, B, C, D, E, F and G are cousins. D is thrice as old as A. Further, C is as many years younger to B, as G to E and E to D. The average age of D and G is 16 years; the average age of A and E is 11 years; the average of B and C is also 11 years. B and C have equal weight. A's weight is 10 kg less than that of B; D is 4 kg heavier than E; E is 4 kg heavier than F; F is 4 kg heavier than G. Further, D has age-weight ratio of 9:20, where age is in years and weight in kg; A has age-weight ratio of 2:5. Moreover, none of them is more than 40kg.

- 142.** What is D's age (in years)?

[CDS-2023 (2)]

- (a) 15 (b) 16
 (c) 17 (d) 18

- 143.** What is the average age (in years) of B, C, D, E and G?

[CDS-2023 (2)]

- (a) 12 (b) 13
 (c) 14 (d) 15

- 144.** What is the difference between the weights (in kg) of G and C?

[CDS-2023 (2)]

- (a) 4 (b) 3
 (c) 2 (d) 1

- 145.** What is the average weight (in kg) of A, B, C, D, E, F and G?

[CDS-2023 (2)]

- (a) 201/7 (b) 197/7
 (c) 30 (d) 32

- 146.** Consider the following statements:

1. The age of F cannot be determined due to insufficient data
2. The average weight of D and F is equal to weight of E
3. The weight difference is maximum for D and A

Which of the statements given above are correct?

[CDS-2023 (2)]

- (a) 1 and 2 only (b) 2 and 3 only
 (c) 1 and 3 only (d) 1, 2 and 3

Consider the following for the next two items that follow:

Mark option (a) if the question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

Mark option (b) if the question can be answered by using either statements alone.

Mark option (c) if the question can be answered by using both the statements together, but cannot be answered using either statement alone.

Mark option (d) if the question cannot be answered even by using both the statements together.

- 147.** Question: What is the other root of the quadratic equation with real coefficients if one of the roots is $\frac{-4 - \sqrt{10}}{2}$?

Statement I:

The product of the roots is $-3/2 (3 + \sqrt{10})$

Statement II:

The sum of roots of quadratic equation is -1.

148. Question: What is the cost of 15 pens, 21 pencils and 18 note books?

Statement I:

The cost of 7 pens, 6 pencils and 5 note books is Rs.200

Statement II:

The of 3 pens, 8 pencils and 7 note books is Rs.210

149. The combined age of a man and his wife is 6 times the combined age of their children. Two years ago their combined age of their children; and six years later their combined age will be 3 times the combined age of their children. How many children do they have if each child is at least 2 years old?

[CDS-2023 (2)]

- (a) 2 (b) 3
(c) 4 (d) 5

150. How many real roots does the equation $\sqrt{x+9} = x-3$ have?

[CDS-2023 (2)]

- (a) only one (b) only two
(c) only three (d) none

151. If $3^{x-1} + 3^{3-x} = 6$, then what is $2^{x-1} + 2^{3-x}$ equal to?

[CDS-2023 (2)]

- (a) 4 (b) 3
(c) 2 (d) 1

Consider the following for the next two (02) items that follow:

Let $\frac{(x-a)(x-b)}{(x-m)(x-nb)} = \frac{(x+a)(x+b)}{(x+ma)(x+nb)}$; $m, a, b > 0$.

152. What is $\frac{x^2+ab}{x^2+m^2ab}$ equal to?

[CDS-2023 (2)]

- (a) $-\frac{1}{m^2}$ (b) $\frac{1}{m^2}$
(c) $\frac{2}{m}$ (d) $\frac{1}{m}$

153. What is x equal to?

[CDS-2023 (2)]

- (a) $\pm\sqrt{mab}$ (b) $\pm\sqrt{ab}$
(c) $\pm\sqrt{2mab}$ (d) $\pm\sqrt{2ab}$

Consider the following for the next two (02) items that follow:

The total monthly electricity bill for a house consists of the sum of two parts, one part is proportional to number of rooms and the other part is proportional to number of

units consumed. Rs.400 is the monthly electricity bill for the a house with 8 rooms and consuming 240 units and Rs.320 is the monthly electricity bill for a house with 6 rooms and consuming 200 units.

154. What is the monthly electricity bill for a house with m rooms and consuming n units?

[CDS-2023 (2)]

- (a) Rs.(40m + n) (b) Rs.(20m + n)
(c) Rs. $\frac{(40m+n)}{2}$ (d) Rs. $\frac{(30m+n)}{2}$

155. What is the monthly electricity bill for a house with 7 rooms consuming 300 units?

[CDS-2023 (2)]

- (a) Rs.500 (b) Rs.440
(c) Rs.340 (d) Rs.300

Consider the following for the next two (02) items that follow:

A quadratic equation is given by $(a+b+c)x^2 - (2a+2b)x + (a+b-c)=0$; where a, b and c are real and distinct.

156. What are the roots of the equation?

[CDS-2023 (2)]

- (a) $1, \frac{(a+b-c)}{(a+b+c)}$ (b) $1, \frac{(a-b+c)}{(a+b+c)}$
(c) $-1, \frac{(-a-c+c)}{(a+b+c)}$ (d) $-1, \frac{(a+b-c)}{(a+b+c)}$

157. Consider the following statements:

- 1.One of the roots of the equation is always less than 1 if a, b and c are all positive
 - 2.One of two roots of the equation is always negative if a, b and c are all negative.
- Which of the statements given above is/are correct?

[CDS-2023 (2)]

- (a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

158. If the equation $x \cos \theta = x^2 + p$ has a real solution for every θ where $0 \leq \theta \leq \frac{\pi}{4}$, then which one of the following is correct?

[CDS-2024 (1)]

- (a) $p = 1/8$ (b) $p \leq 1/8$
(c) $p \geq 1/8$ (d) $p \leq 1/4$

159. What is the greatest value of k for which $2x^2 - 4x + k = 0$ has real roots?

[CDS-2024 (1)]

- (a) 1 (b) 2

(c) 3

(d) 4

160. If the sum and product of the roots of a quadratic equation are 2 and -100 respectively, then which one of the following is correct?

[CDS-2024 (1)]

(a) There are infinitely many such equations having differing roots

(b) There is only one such equation which is $x^2 + 2x - 100 = 0$

(c) There is only one such equation which is $x^2 - 2x - 100 = 0$

(d) There is no such equation

ANSWER KEY

1.	a	2.	d	3.	b	4.	d	5.	a	6.	a	7.	a	8.	d	9.	b	10.	d
11.	b	12.	c	13.	b	14.	d	15.	a	16.	a	17.	c	18.	a	19.	a	20.	d
21.	d	22.	c	23.	a	24.	b	25.	a	26.	b	27.	b	28.	c	29.	a	30.	c
31.	a	32.	d	33.	c	34.	c	35.	c	36.	b	37.	d	38.	c	39.	a	40.	c
41.	b	42.	a	43.	a	44.	b	45.	a	46.	a	47.	c	48.	d	49.	b	50.	c
51.	d	52.	b	53.	b	54.	b	55.	c	56.	a	57.	a	58.	c	59.	a	60.	b
61.	d	62.	a	63.	b	64.	d	65.	a	66.	c	67.	c	68.	c	69.	c	70.	a
71.	b	72.	c	73.	a	74.	b	75.	b	76.	c	77.	c	78.	b	79.	b	80.	c
81.	b	82.	b	83.	a	84.	b	85.	d	86.	c	87.	b	88.	d	89.	c	90.	b
91.	a	92.	b	93.	c	94.	b	95.	a	96.	d	97.	d	98.	c	99.	a	100.	d
101.	b	102.	a	103.	c	104.	b	105.	c	106.	b	107.	d	108.	c	109.	d	110.	b
111.	d	112.	b	113.	b	114.	b	115.	b	116.	c	117.	a	118.	c	119.	c	120.	c
121.	d	122.	b	123.	b	124.	a	125.	b	126.	c	127.	c	128.	a	129.	b	130.	c
131.	b	132.	a	133.	d	134.	a	135.	a	136.	c	137.	a	138.	b	139.	b	140.	a
141.	d	142.	d	143.	c	144.	b	145.	a	146.	a	147.	b	148.	c	149.	a	150.	b
151.	a	152.	d	153.	a	154.	b	155.	b	156.	a	157.	a	158.	b	159.	b	160.	c

CAPF PYQ

1. A number is 124 more than its one-third. What is that number?

[CAPF 2022]

(a) 194
(c) 189

(b) 180
(d) 186

2. A test consists of 25 MCQs. Each correct answer gives +4 marks and incorrect answer gives -1 mark. If a candidate scores 74 marks, then how many questions were left un-attempted?

(a) 4
(c) 5

(b) 3
(d) 9

3. A person has a total of 100 coins consisting of ₹2 and ₹5 coins. If the total value of the coins is ₹320, then the number of ₹2 coins is:

[CAPF 2022]

(a) 40
(c) 60

(b) 50
(d) 70

ANSWER KEY

1.	d	2.	a	3.	c														
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